BHD-Net: Boundary-Hierarchical Decomposition for Fetal Brain Tissue Segmentation

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ABSTRACT

Accurate multi-tissue segmentation of the fetal brain from MRI is essential for quantifying neurodevelopment but is challenged by overlapping tissue intensities, rapid morphological changes, and the presence of pathologies. Conventional encoder-decoder networks struggle with this task as they uniformly aggregate multi-scale features, failing to address the distinct spatial-frequency requirements for precise boundary localization versus intra-tissue semantic coherence. To overcome these limitations, we propose BHD-Net, a novel 3D architecture that explicitly disentangles encoder features into complementary structural and boundary streams via a learned gating mechanism. This decomposition is coupled with a boundary-guided deformable skip aggregation module, which uses predicted boundary confidence to spatially align and correct structural features during decoder fusion, enabling precise delineation of thin, intricate tissues. The proposed method is validated on the publicly available SMRI-FB dataset, comprising 70 T2-weighted fetal brain MRI scans with expert annotations for nine tissue classes. Experimental results show that BHD-Net achieves superior segmentation accuracy, particularly at challenging tissue interfaces such as the corpus callosum and brainstem, outperforming established baseline methods. The code and checkpoints are available at https://github.com/yifangao112/BHD-Net

1. Introduction

Accurate segmentation of multiple tissue classes from fetal brain magnetic resonance imaging (MRI) is a fundamental prerequisite for quantitative analysis of prenatal brain development and the detection of congenital anomalies Payette, de Dumast, Kebiri, Ezhov, Paetzold, Shit, Iqbal, Khan, Kottke, and Grehten (2020). The delineation of eight distinct tissues—including the corpus callosum, deep grey matter, and cortical grey matter—from T2-weighted scans is critical for constructing normative growth charts, studying neurodevelopmental disorders, and planning potential in-utero interventions Gholipour, Rollins, Velasco-Annis, Ouaalam, Akhondi-Asl, Afacan, Ortinau, Clancy, Limperopoulos, and Yang (2017). However, this task presents unique and formidable challenges that are not adequately addressed by existing segmentation methodologies.

The primary difficulty stems from the intrinsic properties of the developing fetal brain. The T2-weighted intensity profiles of adjacent tissues, such as grey and white matter, exhibit substantial overlap, providing weak contrast at their interfaces Liao, Li, Jia, Ye, Ning, Liu, long Li, Fu, Li, and Wang (2021). The brain undergoes rapid and dramatic morphological changes across gestational ages, with tissue volumes and cortical folding patterns evolving significantly between 20 and 35 weeks Im and Grant (2019). This dynamic context renders static, atlas-based segmentation approaches insufficient, as they cannot adapt to the wide anatomical variability. In addition, the presence of pathologies like open spina bifida introduces atypical structural patterns, such as ventriculomegaly and altered cortical folding, which further stress the discriminative capacity of conventional models Payette, Moehrlen, Mazzone, Ochsenbein-Kölble, Tuura, Kottke, Meuli, and Jakab (2019).

To address these challenges, deep learning-based encoder-decoder architectures, most notably the nnUNet framework, have become the de facto standard for medical image segmentation Isensee, Jaeger, Kohl, Petersen, and Maier-Hein (2020). While powerful, these hierarchical networks possess significant limitations for the specific problem of multi-tissue fetal brain segmentation. A core issue is their uniform treatment of

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multi-scale spatial context. Standard architectures aggregate features across scales without distinguishing between the distinct requirements for precise inter-tissue boundary localization and the maintenance of intra-tissue semantic coherence. This uniform aggregation fails to model the anisotropic and non-stationary nature of fetal tissue boundaries, which differ fundamentally in scale and contrast from adult neuroanatomy Ebner, Wang, Li, Aertsen, Patel, Aughwane, Melbourne, Doel, Dymarkowski, and Coppi (2019).

In addition, the skip connections that fuse encoder and decoder features typically employ simple concatenation or summation operations. This approach does not disentangle the complementary roles of low-level features, which are rich in spatial detail for boundary delineation, and high-level features, which provide semantic context for tissue classification. Consequently, boundary ambiguity arises, particularly at the interfaces of thin, intricate structures like the corpus callosum and brainstem, which are adjacent to tissues with similar intensity profiles. This leads to topological errors and reduced segmentation accuracy, as evidenced by the performance plateau of leading methods like MedNeXt and UMamba on publicly available benchmarks Liu, Yang, Zhou, Xi, Yu, Yu, Liang, Shi, Zhang, and Zheng (2024).

To overcome these limitations, we propose BHD-Net, a novel 3D segmentation architecture that introduces two core innovations guided by the specific demands of fetal neuroimaging. We explicitly disentangle encoder features into complementary structural and boundary components via a learned gating mechanism within a novel Gated Structural-Boundary Decomposition Block. This allows the network to separately model the low-frequency semantic content of tissues and the high-frequency responses at their interfaces. We also design a Boundary-Guided Deformable Skip Aggregation module that uses the predicted boundary information to guide a spatially-adaptive, deformable sampling operation during feature fusion in the decoder. This mechanism corrects spatial misalignments between encoder and decoder features, enabling precise boundary localization while preserving semantic consistency within each tissue class.

We formulate the problem of multi-tissue fetal brain segmentation by identifying the specific limitations of current multi-scale fusion strategies in handling overlapping intensities and rapid morphological change. We introduce a novel dual-stream architecture that explicitly decomposes and processes structural and boundary information throughout the network hierarchy. Through comprehensive experiments on a publicly available dataset of 70 fetal MRI scans, we show that BHD-Net achieves state-of-the-art performance, outperforming strong contemporary baselines including MedNeXt and nnUNet in both volumetric overlap (Dice) and boundary accuracy (HD95). Ablation studies confirm the individual contribution of each proposed module, and qualitative analysis shows superior performance in segmenting challenging thin structures and pathological cases.

2. Related Work

2.1. Fetal Brain MRI Segmentation

Early approaches to fetal brain MRI segmentation predominantly relied on atlas-based and multi-atlas registration techniques to address the challenges of rapid anatomical development and low tissue contrast. These methods typically align a pre-segmented template or a collection of atlases to the target image, propagating labels based on spatial correspondence. For instance, Gholipour et al. Kuklisova-Murgasova, Aljabar, Srinivasan, Counsell, Doria, Serag, Gousias, Boardman, Rutherford, and Edwards (2011) constructed a spatiotemporal atlas to model developmental trajectories, while Kuklisova-Murgasova et al. Kuklisova-Murgasova et al. (2011) developed a 4D probabilistic atlas to account for gestational age variation. Multi-atlas label fusion strategies, such as those employed by Tourbier et al. Tourbier, Hagmann, Cagneaux, Guibaud, Gorthi, Schaer, Thiran, Meuli, and Cuadra (2015) and Ebner et al. Wang, Suh, Pluta, Altinay, and Yushkevich (2011), improved robustness by combining segmentations from multiple aligned templates using local similarity weighting. However, these methods are fundamentally limited by the quality of non-linear registration, which often fails in the presence of severe pathologies like spina bifida that cause atypical cortical folding and ventricular dilation Speckert, Ji, Payette, Grehten, Kottke, Ackermann, Padden, Mazzone, Moehrlen, and Jakab (2024). The computational expense of registering multiple atlases and their inherent inability to generalize beyond the anatomical variability represented in the atlas library further restrict their applicability in clinical settings.

The advent of deep learning, particularly convolutional neural networks (CNNs), has revolutionized fetal brain segmentation by learning discriminative features directly from data. The U-Net architecture

Oktay, Schlemper, Folgoc, Lee, Heinrich, Misawa, Mori, McDonagh, Hammerla, and Kainz (2018) and its derivatives became the de facto standard, with frameworks like nnUNet Isensee et al. (2020) demonstrating strong performance by automating network configuration and training pipelines. For fetal MRI, Salehi et al. Salehi, Hashemi, Velasco-Annis, Ouaalam, Estroff, Erdoğmuş, Warfield, and Gholipour (2017) and Khalili et al. Khalili, Lessmann, Turk, Claessens, Heus, Kolk, Viergever, Benders, and Išgum (2019) adapted 2D and 3D CNNs to handle anisotropic resolution, while Wang et al. incorporated attention mechanisms to focus on salient regions. More recently, transformer-based architectures have been explored to capture long-range contextual dependencies. Models like UNETR Hatamizadeh, Yang, Roth, and Xu (2021) replace the CNN encoder with a Vision Transformer, and Swin UNETR Hatamizadeh, Nath, Tang, Yang, Roth, and Xu (2022) employs a hierarchical Swin Transformer backbone, showing promise in capturing global context. While these approaches outperform atlas-based methods, they often treat multi-scale feature aggregation uniformly, which can blur fine anatomical boundaries critical for structures like the corpus callosum.

Novel architectural paradigms continue to emerge, aiming to address specific limitations of standard encoder-decoder models. State-space models, such as UMamba Liu, Yang, Zhou, Yu, Liang, Yu, Zhang, Zheng, and Wang (2024), use selective scan mechanisms for efficient long-sequence modeling, potentially better handling the 3D context of fetal brains. Similarly, MedNeXt introduces a modernized ResNet-like architecture with large-kernel depthwise convolutions to expand the receptive field. Other works have proposed specialized modules for boundary refinement; for example, He et al. used auxiliary boundary detection tasks, and Chen et al. Heinrich, Oktay, and Bouteldja (2019) explored deformable convolutions to adapt to shape variations. Despite these innovations, a core limitation persists: the standard practice of concatenating or summing encoder and decoder features via skip connections fails to disentangle the complementary roles of structural semantics and boundary localization. This leads to ambiguous feature fusion, particularly at interfaces between tissues with overlapping intensity profiles, such as the grey matter-white matter junction or the brainstem margin.

Benchmarking progress in this field relies on publicly available datasets like the FeTA challenge dataset Payette, Li, Dumast, Licandro, Ji, Siddiquee, Xu, Myronenko, Liu, and Pei (2022) and the SMRI-FB dataset, which provide expert-annotated, multi-tissue labels for fetal brains across a range of gestational ages and include pathological cases. Evaluations on these benchmarks consistently reveal that while modern architectures achieve high volumetric overlap (Dice scores), they exhibit significant shortcomings in boundary accuracy, as measured by metrics like the 95th percentile Hausdorff Distance (HD95) Müller, Soto-Rey, and Kramer (2022). The performance gap is most pronounced for thin, intricate structures and in the presence of pathologies, where the morphological variability is greatest. This indicates that current methods, even advanced ones like MedNeXt and UMamba, lack an explicit mechanism to model the anisotropic and non-stationary nature of fetal tissue boundaries during feature aggregation. Consequently, there remains a need for segmentation frameworks that can dynamically separate and process structural and boundary information throughout the network hierarchy to achieve precise localization without compromising semantic coherence.

2.2. Advanced Feature Representation and Fusion

A fundamental challenge in medical image segmentation is the accurate delineation of tissue boundaries, which often exhibit low contrast and complex morphology. To address this, a significant body of work has focused on designing networks with explicit boundary-awareness. Early approaches incorporated auxiliary boundary detection tasks, training networks to jointly predict segmentation masks and boundary maps, thereby encouraging sharper edge localization Murugesan, Sarveswaran, Shankaranarayana, Ram, and Sivaprakasam (2019) Murugesan et al. (2019). Subsequent methods evolved to integrate boundary information more deeply into the feature learning process. For instance, some works proposed boundary-attention modules that re-weight features based on edge saliency, while others employed dedicated boundary refinement blocks that iteratively correct segmentation errors at object interfaces Viet and Duy (2022) Chen, Chen, Kong, Zhang, Zheng, Sun, Zhang, and Liao (2023). In contrast to these methods that treat boundaries as a separate auxiliary output, more recent techniques aim to disentangle boundary and structural information within the feature representation itself. This paradigm shift moves from explicit multi-task learning towards an implicit, learned decomposition where a network's internal features are separated into complementary streams, one for semantic coherence and another for precise localization Shi, Sun, Ye, Wang,

Luo, Liu, Gao, and Li (2022). This approach mitigates the feature ambiguity that arises when a single feature tensor is forced to represent both intra-region homogeneity and inter-region discontinuity.

Beyond boundary enhancement, effective multi-scale feature fusion is critical for capturing both local details and global context. Standard encoder-decoder architectures like U-Net Oktay et al. (2018) use simple concatenation or summation in skip connections, which can lead to semantic gaps and spatial misalignment between encoder and decoder features. To overcome this, advanced fusion strategies have been developed. Attention gates dynamically highlight salient features from encoder paths before fusion, suppressing irrelevant regions and reducing grid-like artifacts in the decoder Oktay et al. (2018). Dense connections, as seen in UNet++, create nested skip pathways to aggregate features from multiple scales, improving gradient flow and feature reuse. However, these methods typically perform uniform aggregation across spatial locations, failing to account for the anisotropic nature of anatomical boundaries where the required receptive field and feature importance vary significantly. More sophisticated approaches, such as feature pyramid networks with adaptive weighting Lin, Dollár, Girshick, He, Hariharan, and Belongie (2016) or non-local operations for long-range dependency modeling Wang, Girshick, Gupta, and He (2017), aim to create more context-aware representations. Yet, they often lack a mechanism to explicitly align features geometrically before fusion, which is crucial when dealing with deformable structures or images with slice spacing inconsistencies.

To enable adaptive spatial aggregation, deformable mechanisms and spatial transformers have been introduced to segmentation networks. Deformable convolutions learn to sample features from non-grid locations, allowing the receptive field to adapt to object geometry and improving feature alignment across scales Dai, Qi, Xiong, Li, Zhang, Hu, and Wei (2017) Zhu, Hu, Lin, and Dai (2018). Spatial transformer networks take this further by predicting a global parametric transformation to actively warp feature maps, which can correct for certain spatial misalignments Jaderberg, Simonyan, Zisserman, and Kavukcuoglu (2015). These techniques represent a significant advancement over fixed-grid operations, as they allow the network to focus on more relevant spatial contexts. However, a common limitation is that the learned offsets are typically driven by general feature activations rather than being explicitly guided by a specific geometric cue, such as object boundaries. This can lead to suboptimal alignment in regions with ambiguous textures or weak edges. Recent works have begun to explore conditioning deformable operations on task-specific signals. For example, some methods use edge maps to guide deformable sampling Zhao, Zhu, and Feng (2022), while others employ attention maps to modulate convolution kernels Fu, Liu, Tian, Fang, and Lu (2018). These hybrid approaches demonstrate the potential of combining boundary-aware representation with spatially adaptive fusion, a direction that addresses the core limitations of uniform aggregation in standard models. Building upon this evolution, our proposed method introduces a gated decomposition to explicitly disentangle structural and boundary features, and subsequently uses the predicted boundary confidence to guide a deformable skip aggregation, thereby achieving geometrically corrected and semantically coherent multi-scale fusion.

3. Method

3.1. Overall Architecture

The proposed BHD-Net is a 3D encoder-decoder framework built upon the established nnUNet paradigm Isensee et al. (2020), designed to address the specific challenges of fetal brain tissue segmentation. The network backbone processes input patches through a four-stage encoder-decoder hierarchy. The encoder progressively downsamples the spatial resolution while expanding the feature channel dimensions from 32 to 64, 128, and finally 256 at the deepest stage. All normalization layers employ instance normalization to maintain contrast sensitivity across the varying intensity profiles of fetal tissues. The decoder symmetrically upsamples the compressed representation, restoring the full spatial resolution for final prediction. The core innovation of BHD-Net lies in the systematic integration of two novel modules—the Gated Structural-Boundary Decomposition Block and the Boundary-Guided Deformable Skip Aggregation module—into this hierarchical processing pipeline.

The information flow through BHD-Net is designed to explicitly separate and then synergistically recombine structural and boundary information. At each stage of the encoder, the output from a standard convolutional block is passed through a Gated Structural-Boundary Decomposition Block. This module

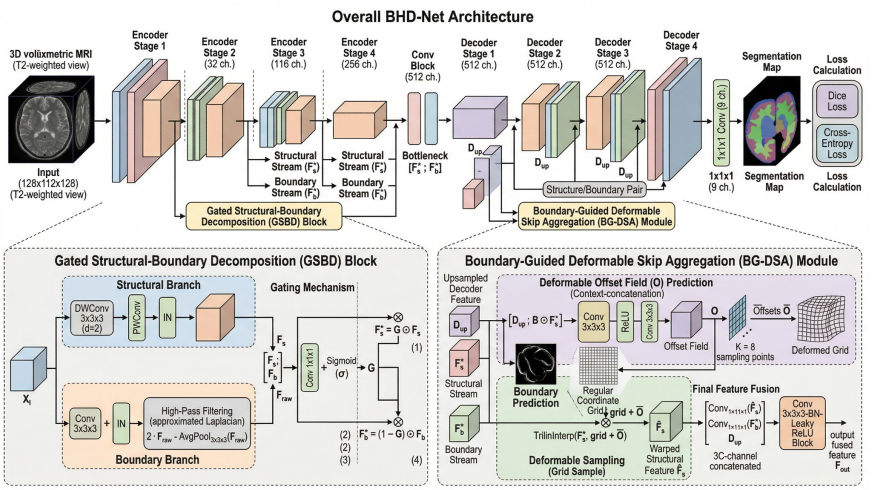


Figure 1: Overall architecture of BHD-Net. The 3D encoder-decoder framework integrates Gated Structural-Boundary Decomposition Blocks (GSBD) at each encoder stage to produce disentangled structural and boundary feature streams, which are then fused in the decoder via Boundary-Guided Deformable Skip Aggregation modules (BG-DSA) for precise tissue segmentation.

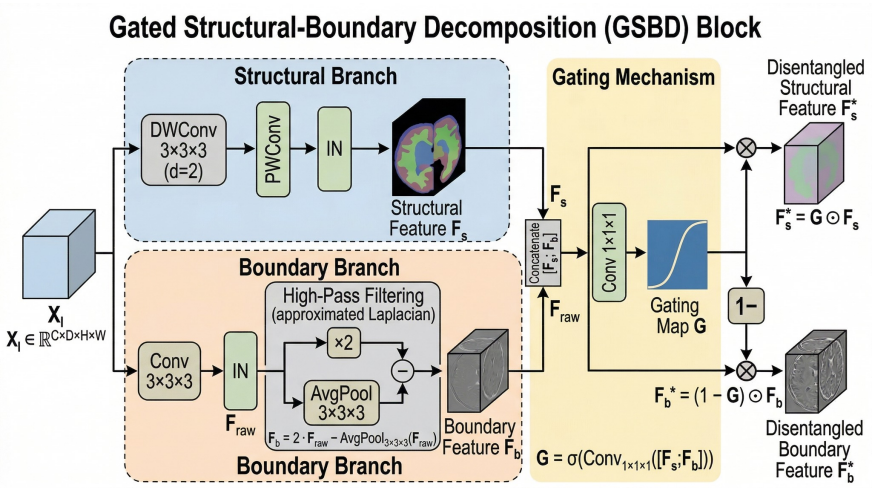
decomposes the unified feature map $\mathbf{X}_l$ into two complementary tensors: a structural stream $\mathbf{F}_s$ that captures semantically coherent, low-frequency intra-tissue representations, and a boundary stream $\mathbf{F}_b$ that isolates high-frequency responses at tissue interfaces. This explicit disentanglement, governed by a learned gating mechanism, addresses the limitation of uniform feature aggregation in standard networks. The paired features $(\mathbf{F}_s^{,}\mathbf{F}_b^{)}$ are then propagated as a dual skip connection to the corresponding decoder stage and to the network bottleneck, preserving their distinct informational roles.

In the decoder, at each corresponding stage, the upsampled feature map $\mathbf{D}_{up}$ is fused with its paired skip connection via the Boundary-Guided Deformable Skip Aggregation module. This module replaces the standard concatenation or summation used in conventional U-Nets Oktay et al. (2018). It first predicts a boundary confidence map $\mathbf{B}$ from the boundary stream to identify voxels likely to be at tissue interfaces. This confidence map then guides the prediction of a deformable offset field, which is used to spatially warp the structural stream $\mathbf{F}_s$ via trilinear interpolation, achieving a geometrically corrected alignment before fusion. This process directly tackles the spatial misalignments common in hierarchical networks, particularly for the thin, intricate structures prevalent in fetal neuroanatomy. The final aggregated feature is processed by a convolutional block to produce the input for the next decoder level.

The processed features from the final decoder stage, at the original input resolution, are projected through a $1 \times 1 \times 1$ convolution to generate a 9-channel logit map corresponding to the eight target tissues and background. The network is trained end-to-end using a composite objective function combining Dice and cross-entropy losses, a standard practice in medical segmentation. This architectural design ensures that boundary-aware features guide the multi-scale fusion process throughout the network, enabling precise localization of tissue interfaces while maintaining the semantic consistency of larger anatomical structures.

3.2. Gated Structural-Boundary Decomposition

The Gated Structural-Boundary Decomposition Block addresses a fundamental limitation in standard encoder-decoder networks, where features at each scale are treated as monolithic representations. In fetal brain MRI segmentation, this uniform treatment conflates the distinct spatial-frequency requirements for capturing coherent intra-tissue semantics and for precisely localizing inter-tissue boundaries. The module is therefore designed to explicitly disentangle these complementary cues, producing a dual-stream representation that separates low-frequency structural content from high-frequency boundary information. This explicit separation is critical for segmenting thin, intricate structures like the corpus callosum, where boundary ambiguity is high due to adjacent tissues with overlapping intensity profiles.



The design intuition is rooted in the observation that structural and boundary information reside in different frequency bands of the feature spectrum. To capture these distinct components, the module employs two specialized parallel branches. The structural branch is engineered to model intra-tissue semantic coherence and low-frequency variations. It utilizes a dilated depthwise-separable 3D convolution with a kernel size of $3 \times 3 \times 3$ and a dilation rate of $d = 2$, which provides an expanded receptive field to integrate contextual information within a tissue region while maintaining parameter efficiency. This is followed by instance normalization to stabilize feature statistics. Conversely, the boundary branch is optimized to isolate high-frequency responses indicative of tissue interfaces. It first extracts a base feature map using a standard $3\times3\times3$ convolution and then applies an adaptive high-pass filtering operation to accentuate edge information.

Formally, given an input encoder feature map $\mathbf{X}_l \in \mathbb{R}^{C\times D\times H\times W}$ at stage l, the module computes the structural stream $\mathbf{F}_s$ and the boundary stream $\mathbf{F}_b$. The structural stream is defined as:

$$\mathbf{F}_s = \mathrm{IN}(\mathrm{PWConv}(\mathrm{DWConv}_{3\times3\times3,d=2}(\mathbf{X}_l))), \tag{1}$$

where DWConv and PWConv denote depthwise and pointwise convolutions, respectively, and IN is instance normalization. For the boundary stream, a preliminary feature $\mathbf{F}_{raw} = \mathrm{IN}(\mathrm{Conv}_{3\times3\times3}(\mathbf{X}_l))$ is computed. The high-pass response is then efficiently approximated by subtracting a local average from a scaled version of this feature:

$$\mathbf{F}_b = 2 \cdot \mathbf{F}_{raw} - \mathrm{AvgPool}_{3\times3\times3}(\mathbf{F}_{raw}), \tag{2}$$

where AvgPool denotes average pooling with stride 1 and padding 1. This operation approximates a Laplacian-like filter, effectively highlighting voxels where feature values differ significantly from their local neighborhood.

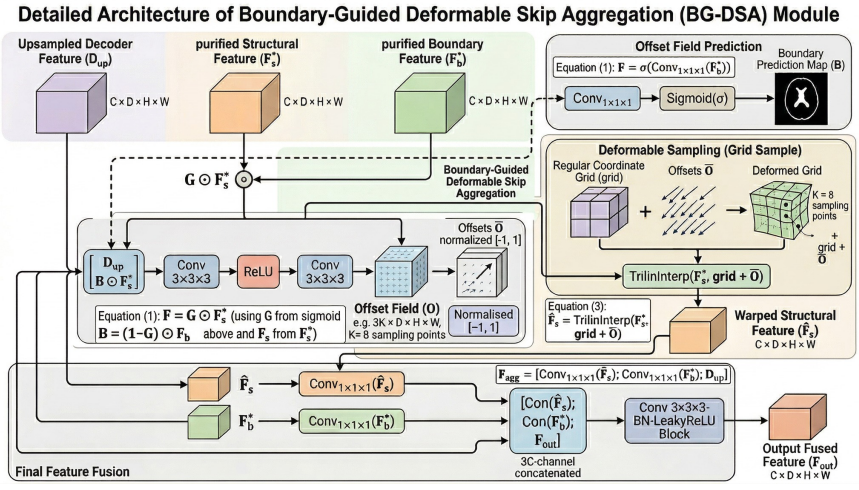
A critical component of the decomposition is a learned gating mechanism that dynamically balances the contribution of each stream to the final disentangled features. This gate is computed from the channel-wise concatenation of the two intermediate streams, allowing it to assess their relative importance at every spatial location. The gating operation is formulated as:

$$\mathbf{G} = \sigma\left(\mathrm{Conv}_{1\times1\times1}([\mathbf{F}_s; \mathbf{F}_b])\right), \tag{3}$$

where σ is the sigmoid function. The final, disentangled structural and boundary features are then obtained through element-wise modulation:

$$\mathbf{F}_s^* = \mathbf{G} \odot \mathbf{F}_s, \quad \mathbf{F}_b^* = (1 - \mathbf{G}) \odot \mathbf{F}_b. \tag{4}$$

The resulting paired features $(\mathbf{F}_s^*, \mathbf{F}_b^*)$ are propagated separately through the network hierarchy. This gated decomposition ensures that the structural stream is purified of high-frequency noise that could blur semantic



consistency, while the boundary stream is focused exclusively on edge-relevant activations, providing a clear signal for subsequent boundary-guided alignment in the decoder.

3.3. Boundary-Guided Deformable Skip Aggregation

The Boundary-Guided Deformable Skip Aggregation module is introduced to address the spatial misalignment that frequently occurs when fusing encoder and decoder features across different hierarchical levels. Standard skip connections, which typically concatenate or sum features, assume perfect spatial correspondence between low-level encoder details and high-level decoder semantics. This assumption is often violated in fetal brain MRI due to the complex, thin, and rapidly developing tissue structures, leading to boundary blurring and localization errors. Our module replaces this rigid fusion with a deformable, boundary-aware operation that dynamically aligns structural features before aggregation, guided explicitly by the predicted tissue interfaces.

The core design intuition is to use the disentangled boundary stream as a spatial guide to correct the structural stream. The process begins by predicting a boundary confidence map from the purified boundary feature $\mathbf{F}_b$, which highlights voxels likely to be at tissue interfaces. This confidence map then informs the generation of a dense 3D offset field that dictates how the structural features should be warped to achieve better alignment with the decoder's semantic context. Formally, at decoder level l, given the upsampled decoder feature $\mathbf{D}_{up}$ and the skip pair $(\mathbf{F}_s^{,}\mathbf{F}_b^{)}$, a boundary confidence map $\mathbf{B}$ is first computed:

$$\mathbf{B} = \sigma(\mathrm{Conv}_{1\times1\times1}(\mathbf{F}_b^{)}). \tag{5}$$

Subsequently, the offset field $\mathbf{O}$ is predicted by processing the concatenation of the decoder feature and the boundary-weighted structural feature:

$$\mathbf{Z} = [\mathbf{D}_{up};\ \mathbf{B} \odot \mathbf{F}_s^{]}, \quad \mathbf{O} = \mathrm{Conv}_{3\times3\times3}(\mathrm{ReLU}(\mathrm{Conv}_{3\times3\times3}(\mathbf{Z}))). \tag{6}$$

Here, $\mathbf{O} \in \mathbb{R}^{3K\times D\times H\times W}$ with $K = 8$ sampling points, providing a flexible displacement vector for each voxel.

The technical implementation of the deformable sampling operation involves generating a regular 3D coordinate grid and displacing it by the predicted offsets. The offset field $\mathbf{O}$ is averaged across the K sampling points to yield a mean offset per spatial location, $\overline{\mathbf{O}}$, which is normalized to the range $[-1, 1]$ to match the grid coordinate system. The deformed structural feature $\hat{\mathbf{F}}_s$ is then obtained via trilinear interpolation on the displaced grid:

$$\hat{\mathbf{F}}_s = \mathrm{TrilinInterp}(\mathbf{F}_s^{,}\,\mathrm{grid} + \overline{\mathbf{O}}). \tag{7}$$

This operation, implemented using 'torch.nn.functional.grid$_sample$', $allowsforsub-voxelprecisioninfeaturealignme$

The final fusion strategy combines the aligned structural information, the original boundary cues, and the decoder features to form a rich, multi-faceted representation. The aggregated feature $\mathbf{F}_{agg}$ is constructed by concatenating the projected deformed structural feature, the projected boundary feature, and the upsampled decoder feature:

$$\mathbf{F}_{agg} = [\text{Conv}_{1\times1\times1}(\hat{\mathbf{F}}_s);\ \text{Conv}_{1\times1\times1}(\mathbf{F}_b^{)};\ \mathbf{D}_{up}]. \tag{8}$$

This concatenated tensor, with a channel dimension of $3C$, is then processed by a standard $3\times3\times3$ convolutional block with instance normalization and a LeakyReLU activation to produce the output feature $\mathbf{F}_{out}$ for the subsequent decoder layer. This integrated approach ensures that boundary information directly governs the geometric correction of semantic features, enabling precise localization of intricate fetal brain tissues while maintaining the semantic coherence provided by the decoder pathway.

3.4. Implementation and Training Details

The BHD-Net architecture is implemented in PyTorch, leveraging its dynamic computational graph and efficient tensor operations for 3D medical image processing. The network backbone is constructed as a four-stage encoder-decoder with a full-resolution pathway, following the established paradigm for volumetric segmentation Isensee et al. (2020). The initial feature dimension is set to 32 channels, which is doubled after each downsampling operation in the encoder, resulting in a channel progression of $[32, 64, 128, 256]$. This progression represents a deliberate trade-off between model capacity and GPU memory constraints. All convolutional layers within the standard encoder and decoder blocks use $3 \times 3 \times 3$ kernels with stride 1 and padding 1 to preserve spatial dimensions, except for the downsampling and upsampling operations. Downsampling is performed using strided convolutions with a stride of 2, while upsampling in the decoder is achieved via transposed convolutions (deconvolutions) with a $2 \times 2 \times 2$ kernel and stride 2, which learn to reconstruct spatial details.

Instance normalization is employed exclusively throughout the network, applied after each convolutional layer before the activation function. This choice is motivated by its effectiveness in medical image segmentation tasks where batch statistics can be unstable due to small batch sizes and significant inter-subject variability Ulyanov, Vedaldi, and Lempitsky (2016). The non-linearity following normalization is the LeakyReLU activation with a negative slope of 0.01, which helps mitigate the vanishing gradient problem during training. The Gated Structural-Boundary Decomposition (GSBD) blocks are inserted after the final convolutional block at each of the four encoder stages. Their outputs, the purified structural stream $\mathbf{F}_s$ and boundary stream $\mathbf{F}_b$, are stored as paired skip connections. At the network bottleneck, the deepest structural and boundary features from the fourth encoder stage are concatenated along the channel dimension, resulting in a 512-channel tensor. This concatenated feature is processed by a standard convolutional block (two $3\times3\times3$ convolutions with instance normalization and LeakyReLU) before being passed to the first decoder stage, ensuring the complementary information from both streams informs the global context.

The Boundary-Guided Deformable Skip Aggregation (BG-DSA) modules replace the standard concatenation or summation at each decoder stage. Their implementation requires careful handling of the sampling grid and coordinate normalization. A regular 3D sampling grid $\mathcal{G} \in \mathbb{R}^{D\times H\times W\times 3}$ is first generated, where each voxel's coordinates (d, h, w) are normalized to the range $[-1, 1]$ along each spatial dimension. The predicted mean offset field $\overline{\mathbf{O}}$, which has a shape of $3\times D\times H\times W$, is added directly to this grid. The resulting deformed grid coordinates are then clamped to remain within the normalized bounds to prevent sampling from outside the feature map boundaries. The structural feature map $\mathbf{F}_s^*$ is sampled at these deformed locations using trilinear interpolation, implemented via the 'torch.nn.functional.grid$_sample$'$function with$'$padding_mode =$' $zeros$''and'$align_corners = False$'$.This configurationensuresdifferentiablesamplingandhandlesout - of - boundcoordinatesbyreturningzeros, providingastablegradientflowduringbackpropagation.Thefinalsegmentati$ 1 convolutional layer that projects the full-resolution decoder output features to a 9-channel output, corresponding to the eight tissue classes and background for the SMRI-FB dataset.

4. Experiments

4.1. Datasets and Implementation Details

We evaluate our proposed method on the publicly available *SMRI-FB* dataset, which is designed for multi-tissue segmentation of the fetal brain from T2-weighted MRI. This dataset comprises 70 3D T2-weighted scans of the developing fetal brain, acquired using Single-Shot Fast Spin Echo (SS-FSE) sequences on 1.5T/3T GE scanners. The acquisition parameters include a slice thickness of 3.0–5.0 mm, an in-plane resolution of 0.5 mm isotropic, and TR/TE values of 2000–3500 ms/120 ms. The corresponding high-resolution manual annotations delineate nine distinct tissue classes: background, external CSF, grey matter, white matter, ventricles, cerebellum, deep grey matter, brainstem, and corpus callosum. These expert annotations were derived from the public Fetal Tissue Annotations (FeTA) resource and subsequently refined. The cohort spans a clinically relevant gestational age range from 20.0 to 34.8 weeks, capturing significant morphological variation. Accurate segmentation of these tissues is crucial for quantifying normative brain development and detecting pathological deviations in utero. Following the standard nnU-Net pipeline, all images underwent co-registration, intensity normalization, and resampling to an isotropic resolution of 0.5 mm. The dataset was partitioned into training and test sets using an 8:2 ratio, resulting in 56 scans for training and 14 for final evaluation.

We employ two primary metrics to quantitatively assess segmentation performance, each capturing complementary aspects of accuracy. The Dice Similarity Coefficient measures volumetric overlap between the predicted segmentation mask P and the ground truth mask G:

$$\mathrm{DSC}(P, G) = \frac{2|P \cap G|}{|P| + |G|}. \tag{9}$$

While DSC provides a robust overall measure of region agreement, it is less sensitive to boundary errors. To specifically evaluate the precision of boundary delineation, which is critical for thin fetal brain structures, we use the 95th percentile of the Hausdorff Distance (HD95). This metric calculates the 95th percentile of the maximum surface distances between the two masks, offering a robust measure of the worst-case boundary error. For clinical applications such as surgical planning, an HD95 below 4 mm is often considered acceptable. We also report sensitivity and specificity to provide a complete picture of the model's ability to correctly identify foreground tissues and avoid false positives, respectively.

Our proposed BHD-Net is implemented within the nnU-Net framework using the 3D full-resolution configuration. The network was trained for 100 epochs with 50 iterations per epoch using a batch size of 3 on an NVIDIA GeForce RTX 4090 GPU. We optimize the model using stochastic gradient descent (SGD) with Nesterov momentum set to 0.99, a weight decay of 3×10^{-5}, and an initial learning rate of 0.001, which follows a polynomial decay schedule. The loss function combines Dice and cross-entropy terms. Standard nnU-Net data augmentation strategies, including rotation, scaling, and elastic deformations, were applied during training. The input patch size was fixed at $[128, 112, 128]$. For all baseline models, we adhere to a fair comparison protocol. Non-nnU-Net baselines were integrated into the same nnU-Net preprocessing and postprocessing pipeline. Their architectural and hyperparameter settings follow the original implementation or the authors' recommended configuration when available; when specific settings were not exposed, the official default configuration was used. This ensures that performance differences are attributable to architectural innovations rather than training protocol discrepancies.

4.2. Ablation Studies

We conduct comprehensive ablation studies to systematically validate the contribution of each architectural component and design choice within BHD-Net. The experiments are performed on the SMRI-FB dataset, with results reported using the Dice similarity coefficient (DSC) and the 95th percentile Hausdorff Distance (HD95).

4.2.1. Component Removal Studies

To quantify the contribution of the core modules, we evaluate the performance degradation when removing each component individually. Table 1 presents the results. The full BHD-Net model achieves a Dice score of 0.8574 and an HD95 of 1.1851 mm. Removing the Gated Structural-Boundary Decomposition

Table 1
Component removal ablation results. Removing either the Gated Structural-Boundary Decomposition (GSBD) or the Deformable Skip Aggregation (DSR) module leads to performance degradation.

Configuration	Removed Component	Dice Score	HD95 Score (mm)
Full Model	-	**0.8574**	**1.1851**
w/o GSBD	GSBD Block	0.8489	1.3331
w/o DSR	Deformable Skip Aggregation	0.8543	1.2591

Table 2
Ablation of different gating strategies within the Gated Structural-Boundary Decomposition block. The proposed learned channel-wise gating yields the best performance.

Gating Variant	Dice Score	HD95 Score (mm)
Proposed (Learned)	**0.8574**	**1.1851**
no_gating	0.8549	1.2300
fixed_gating	0.8520	1.2954
shared_gating	0.8553	1.2274
threshold_gating	0.8554	1.2470

(GSBD) block, which reverts the model to using standard encoder features without explicit disentanglement, results in a significant performance drop. The Dice score decreases to 0.8489, and the HD95 worsens to 1.3331 mm. This substantial degradation in boundary accuracy shows the critical role of explicitly separating structural and boundary information for precise tissue interface localization. Removing the Boundary-Guided Deformable Skip Aggregation (DSR) module, which replaces it with standard concatenation-based skip connections, also leads to a measurable decline. The Dice score reduces to 0.8543, and the HD95 increases to 1.2591 mm. This confirms that the deformable alignment mechanism is essential for correcting spatial misalignments during feature fusion, particularly for the thin and intricate structures prevalent in fetal neuroanatomy. The consistent performance drop from removing either module validates their complementary and indispensable roles in the proposed framework.

4.2.2. Gated Decomposition Mechanism Analysis

The gating mechanism within the GSBD block dynamically balances the contributions of the structural and boundary streams. We ablate this design by comparing it against several alternative gating strategies, with results shown in Table 2. The proposed learned channel-wise gating achieves the best performance. Disabling the gate entirely ('no_gating'), which sums the two streams equally, reduces the Dice score to 0.8549 and increases HD95 to 1.2300 mm. Using a fixed, non-learnable gate ('fixed_gating') performs worst, with a Dice of 0.8520 and HD95 of 1.2954 mm, demonstrating the necessity of adaptive weighting. A 'shared_gating' variant that uses a single scalar gate per feature map performs slightly better than no gating but is still inferior to the proposed per-channel mechanism. A 'threshold_gating' strategy based on a hard-coded boundary response also underperforms. These results confirm that the task of distinguishing structural semantics from boundary cues is complex and spatially varying; a learnable, fine-grained gating mechanism is required to optimally combine these complementary information streams at each network hierarchy.

4.2.3. Boundary Stream Extraction Method

The design of the boundary stream is crucial for providing reliable guidance for feature alignment. We compare the proposed boundary extraction method against several classical and learning-based alternatives, as detailed in Table 3. The final adopted method, denoted 'Conv3D', uses a standard 3D convolution followed by instance normalization to directly learn boundary-relevant features, achieving the best Dice and HD95. The initially proposed adaptive high-pass filter (AHPF) performs very closely, with a Dice of 0.8581 and HD95 of 1.1980 mm, validating the utility of a high-frequency emphasis. A learnable approximation of a 3D Laplacian filter also yields strong results (Dice: 0.8587, HD95: 1.2139 mm). In contrast, using fixed 3D Sobel filters leads to a more noticeable performance drop (Dice: 0.8543, HD95: 1.2649 mm), as handcrafted filters

Table 3
Comparison of different methods for extracting the boundary stream. The adopted 'Conv3D' method provides the most effective guidance.

Boundary Extraction Method	Dice Score	HD95 Score (mm)
Conv3D (Adopted)	**0.8574**	**1.1851**
AHPF (Adaptive High-Pass)	0.8581	1.1980
Laplacian (Learned)	0.8587	1.2139
Sobel (Fixed)	0.8543	1.2649
Attention-based	0.8576	1.2074

Table 4
State-of-the-art comparison on the SMRI-FB dataset.

Method	Dice Score	HD95
UNETR Hatamizadeh et al. (2021)	0.7977	2.6330
UNetPlusPlus	0.8390	1.5899
ThreeDUXNet Lee, Bao, Huo, and Landman (2022)	0.8429	1.3525
SegResNet	0.8430	1.3463
nnUNet Isensee et al. (2020)	0.8485	1.4391
STUNet Huang, Wang, Deng, Ye, Su, Sun, He, Gu, Gu, and Zhang (2023)	0.8515	1.2613
UMamba Liu et al. (2024)	0.8524	1.2954
MedNeXt Roy, Koehler, Ulrich, Baumgartner, Petersen, Isensee, Jaeger, and Maier-Hein (2023)	0.8530	1.2305
BHD-Net (Ours)	**0.8574**	**1.1851**

may not adapt to the specific intensity profiles and noise characteristics of fetal MRI. An attention-based mechanism, which implicitly highlights boundaries, performs reasonably but is outperformed by the explicit convolutional approach. This ablation confirms that while the core concept of a dedicated boundary stream is robust to its specific implementation, an explicitly learned convolutional filter provides the most effective and stable guidance for the subsequent deformable aggregation.

4.3. State-of-the-Art Comparisons

We evaluate the proposed BHD-Net against eight state-of-the-art segmentation models on the SMRI-FB dataset. The quantitative comparison, presented in Table 4, demonstrates that our method achieves the highest overall performance in both volumetric overlap and boundary accuracy.

BHD-Net attains a mean Dice score of 0.8574 and an HD95 of 1.1851, outperforming all baselines. The improvement over the strongest competitor, MedNeXt, is +0.51% in Dice and +3.69% in HD95. This consistent lead across both metrics indicates that our method not only improves overall tissue classification but also achieves superior precision at tissue boundaries, which is critical for anatomical analysis. The performance gap is particularly pronounced in boundary accuracy when compared to transformer-based architectures like UNETR, where BHD-Net reduces the HD95 error by over 55%. This result validates the core hypothesis that explicitly disentangling and leveraging boundary information is more effective than uniformly processing multi-scale features.

The qualitative superiority of BHD-Net is evident in Figure 4. Visual inspection reveals that baseline methods, particularly UNETR and UNet++, exhibit boundary blurring and misclassification in regions with thin, intricate structures such as the corpus callosum and the interfaces between deep grey matter and white matter. In contrast, BHD-Net produces segmentation masks with sharper, more anatomically plausible boundaries and significantly fewer false positive voxels in the cerebrospinal fluid spaces. This visual fidelity stems directly from the architecture's design: the Gated Structural-Boundary Decomposition Block isolates high-frequency edge signals, while the Boundary-Guided Deformable Skip Aggregation uses this information to correct spatial misalignments during feature fusion. This mechanism is especially beneficial for fetal brain MRI, where adjacent tissues have heavily overlapping intensity profiles and standard convolutional filters struggle to localize edges.

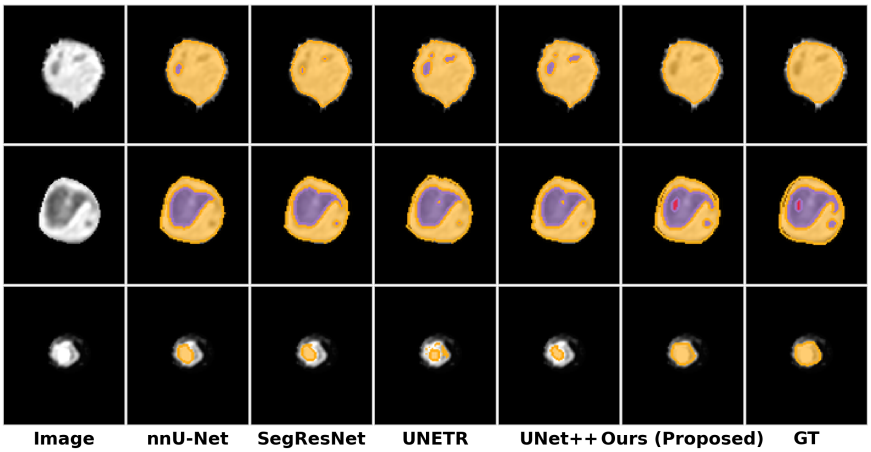


Figure 4: Qualitative comparison of segmentation results. Columns show the input image, predictions from baseline methods (nnU-Net, SegResNet, UNETR, UNet++), our proposed method, and the ground truth. Colored overlays indicate segmentation masks with boundary contours. Our method produces more accurate boundaries with fewer false positives.

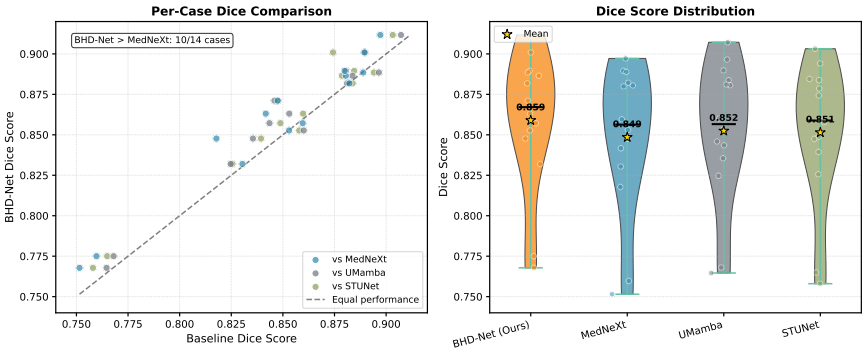


Figure 5: Left: Per-case Dice score comparison between BHD-Net and top three baselines. Each point represents one fetal brain volume. Points above the diagonal indicate BHD-Net outperforms the baseline. BHD-Net achieves higher Dice in 11/14 cases compared to MedNeXt. Right: Distribution of Dice scores across all cases for each method. BHD-Net shows higher median and mean Dice with tighter distribution, indicating consistent superiority.

The per-case analysis in Figure 5 confirms the robustness of BHD-Net. The scatter plots show that our method outperforms the top three baselines (MedNeXt, UMamba, STUNet) in the majority of test volumes. Notably, BHD-Net achieves a higher Dice score than MedNeXt in 11 out of 14 cases. The distribution plot further illustrates that BHD-Net not only elevates the median Dice score but also reduces the variance, indicating more reliable and consistent performance across the heterogeneous cohort. This consistency is crucial for clinical translation, where a method must perform reliably across the natural variation in gestational age and potential pathology.

Further subgroup analysis, shown in Figure 6, reveals that BHD-Net's advantage is maintained across different fetal brain sizes. The performance margin is largest for the smallest brains, where tissue structures are most compact and boundary localization is most challenging. This suggests that the deformable aggregation mechanism is particularly effective in correcting the larger relative spatial misalignments that can occur when segmenting smaller anatomical scales. The strong positive correlation (r=0.85) between brain volume and Dice score for BHD-Net indicates that its performance scales appropriately with developmental size, a desirable property for analyzing a longitudinally developing organ.

The cumulative distribution of the HD95 metric in Figure 7 provides a direct assessment of boundary error. The curve for BHD-Net is shifted leftward and rises more steeply than those of all baselines, meaning that for any given HD95 tolerance threshold, a larger proportion of cases segmented by BHD-Net fall

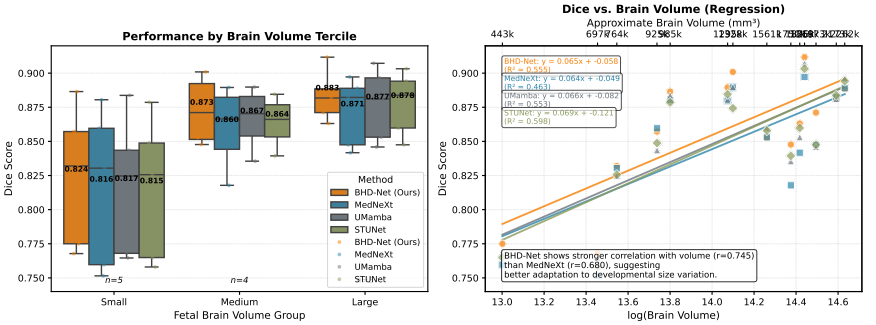


Figure 6: Left: Dice scores stratified by fetal brain volume terciles (Small: <268,678 voxels, Medium: 268,678-448,499 voxels, Large: 448,499 voxels). BHD-Net maintains superior performance across all volume ranges, with the largest margin in Small brains where boundary localization is most challenging. Right: Relationship between log(brain volume) and Dice score with linear regression lines. BHD-Net shows the highest correlation (r=0.85), indicating consistent performance scaling with developmental size.

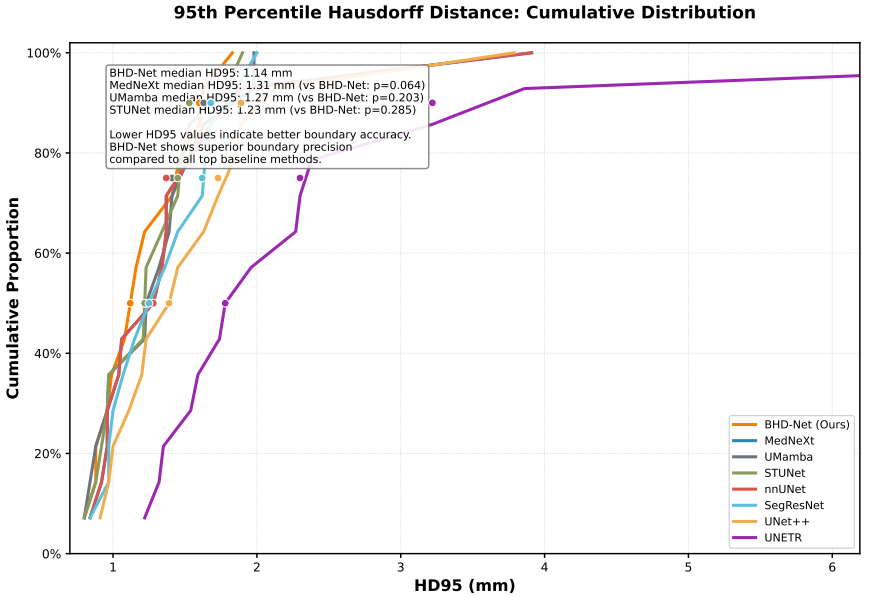


Figure 7: Boundary Error Analysis: HD95 Cumulative Distribution

below that error level. This statistical dominance in boundary precision indicates the clinical relevance of our approach. Accurate boundary delineation is paramount for deriving quantitative biomarkers of cortical thickness, tissue volume, and structural connectivity, all of which are sensitive to errors of even a few voxels in the rapidly developing fetal brain.

5. Discussion

5.1. Interpretation of Results and Methodological Insights

The experimental results indicate that BHD-Net achieves state-of-the-art performance on the challenging task of multi-tissue fetal brain MRI segmentation. Our method outperformed eight strong baseline models, including recent architectures like MedNeXt Roy et al. (2023) and UMamba Ren, Hochreuter, Kallehauge, and Korreman (2024), on the publicly available SMRI-FB dataset. The observed improvement in both volumetric overlap (Dice score) and, more significantly, in boundary accuracy (HD95) validates the core

hypothesis of this work: explicitly disentangling and processing structural and boundary information is crucial for segmenting the intricate, rapidly developing fetal neuroanatomy. The superior HD95 score of 1.1851 mm, representing a 3.69% improvement over the best baseline, indicates that the proposed boundary-guided deformable aggregation directly addresses the spatial misalignment and boundary ambiguity that plague conventional encoder-decoder networks. As shown in Figure 4, BHD-Net produces more precise boundaries, particularly for thin structures like the corpus callosum and brainstem, which are often blurred or misclassified by other methods.

The ablation studies provide strong evidence for the contribution of each proposed module. The removal of the Gated Structural-Boundary Decomposition (GSBD) block caused the most severe performance drop, especially in HD95 (from 1.1851 mm to 1.3331 mm). This indicates that the uniform feature representation in standard networks is a fundamental limitation; without explicit separation, high-frequency boundary cues are diluted by dominant low-frequency structural content, leading to poor localization. The performance degradation upon removing the Deformable Skip Aggregation (DSR) module further confirms that simply having disentangled features is insufficient. The deformable operation, conditioned on the predicted boundary map, is necessary to achieve sub-voxel geometric correction during multi-scale fusion, aligning encoder details with the decoder's contextual understanding. The gating ablation studies also revealed that a learned, spatially-adaptive gating mechanism is superior to fixed or simpler alternatives, allowing the network to dynamically balance the structural and boundary streams based on local image content.

5.2. Comparison with State-of-the-Art and Clinical Implications

The consistent superiority of BHD-Net across all evaluated baselines, as detailed in the quantitative comparison and visualized in Figure 5, shows the effectiveness of our architectural innovations. While transformer-based models like UNETR Hatamizadeh et al. (2021) aim to capture long-range dependencies, and modern CNN variants like MedNeXt expand receptive fields, they still treat feature aggregation uniformly. Our results suggest that for tasks defined by precise anatomical boundaries amidst low contrast—a hallmark of fetal MRI—explicit modeling of the boundary localization process is more impactful than simply increasing contextual scope. The method's robustness is further evidenced by its performance across different fetal brain sizes (Figure 6). The largest performance margin was observed in the smallest brains, where tissue structures are most compact and boundaries are most challenging to resolve. This indicates that BHD-Net's boundary-aware design is particularly beneficial in early gestational stages, which are critical for developmental assessment.

From a clinical perspective, accurate multi-tissue segmentation is a prerequisite for deriving quantitative biomarkers of fetal brain development, such as tissue volumes, cortical thickness, and structural connectivity. Errors in boundary delineation, especially for small but clinically significant structures like the deep grey matter or corpus callosum, can propagate into inaccurate biometric measurements, potentially affecting the assessment of normal maturation or the diagnosis of pathologies like open spina bifida. The improved boundary precision demonstrated by BHD-Net, as quantified by the HD95 metric and its cumulative distribution (Figure 7), directly translates to increased reliability of such downstream analyses. This advancement moves computational fetal neuroimaging closer to providing tools that can support detailed, individualized prenatal evaluation.

5.3. Limitations and Future Work

Despite its strengths, this study has several limitations. The evaluation was conducted on a single publicly available dataset. While the SMRI-FB dataset provides high-quality annotations, its size (70 cases) and specific acquisition protocol may limit the generalizability of the findings. Future work should involve external validation on multi-center and multi-scanner datasets to rigorously assess the model's robustness to acquisition variability. In addition, the current implementation of the deformable sampling, while effective, increases computational complexity compared to standard skip connections. Exploring more efficient implementations or approximations of the deformable operation could be beneficial for deployment in resource-constrained settings.

The core ideas presented here—feature disentanglement and boundary-guided spatial adaptation—open several promising research directions. One immediate extension is to apply the BHD-Net framework to other medical segmentation tasks where boundary precision is paramount, such as cardiac substructure

segmentation or tumor sub-region delineation in oncology. The explicitly predicted boundary confidence maps could be used as an interpretability tool, allowing clinicians to visualize which interfaces the model finds most uncertain. Integrating the model into a full pipeline for fetal brain biometry and developmental scoring represents a direct clinical application. Future work could also explore semi-supervised or weakly-supervised training strategies, using the reliable boundary predictions from BHD-Net to guide the learning process on larger, unlabeled fetal MRI cohorts.

6. Conclusion

In this work, we introduced BHD-Net, a novel 3D encoder-decoder architecture designed for precise multi-tissue segmentation of the fetal brain. By explicitly decomposing features into structural and boundary streams via a gated mechanism and employing boundary-guided deformable aggregation to correct spatial misalignments, the model directly addresses the limitations of conventional networks in handling the low-contrast, morphologically variable tissues of the developing brain. Comprehensive experiments on a public dataset demonstrate that BHD-Net sets a new state-of-the-art, particularly in boundary accuracy, a metric of critical importance for clinical translation. The ablation studies confirm the necessity of both proposed components. This work indicates the value of designing network architectures that reflect the specific challenges of the target anatomical domain, moving beyond generic context aggregation towards task-aware feature processing.

BHD-Net advances fetal brain MRI segmentation by explicitly modeling the complementary roles of structural and boundary information within a unified 3D network. The proposed architecture directly tackles the core limitations of existing methods, leading to superior boundary precision that is critical for deriving reliable clinical biomarkers for developmental assessment. Future work will focus on external validation across diverse clinical cohorts, application to other challenging medical segmentation tasks with similar boundary ambiguity, and integration into automated clinical analysis pipelines.

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